

AI MODEL DEPLOYMENT & MLOPS

Dockerfile & Kubernetes YAML files:

<https://github.com/AFZAALAHMEDKHAN/LiaPlus-AI-assignment.git>

MODEL WORKING:

```
afzaa@AFZAALAHMEDKHAN MINGW64 -
$ minikube status
w0415 17:15:52.840483 10086 main.go:291] Unable to resolve the current Docker CLI context "default": context "default": context not found: open C:\Users\afzaa\.docker\con
texts\meta\37a8ee1ce19687d132fe29051dca629d164e2c4958ba141d5f4133a33f0688f\meta.json: The system cannot find the path specified.
minikube
type: Control Plane
host: Running
kubelnet: Running
apiserver: Running
kubeconfig: Configured

afzaa@AFZAALAHMEDKHAN MINGW64 -
$ cd Desktop\ai\model\
afzaa@AFZAALAHMEDKHAN MINGW64 ~/Desktop/ai/model
$ kubectl apply -f deployment.yaml
deployment.apps/ml-model-deployment created

afzaa@AFZAALAHMEDKHAN MINGW64 ~/Desktop/ai/model
$ kubectl apply -f service.yaml
service/ml-model-service created

afzaa@AFZAALAHMEDKHAN MINGW64 ~/Desktop/ai/model
$ kubectl get deploy
NAME READY UP-TO-DATE AVAILABLE AGE
ml-model-deployment 0/1 1 0 17s

afzaa@AFZAALAHMEDKHAN MINGW64 ~/Desktop/ai/model
$ kubectl get svc
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
kubernetes ClusterIP 10.96.0.1 <none> 443/TCP 3m51s
ml-model-service NodePort 10.107.90.35 <none> 80:30007/TCP 14s

afzaa@AFZAALAHMEDKHAN MINGW64 ~/Desktop/ai/model
$ kubectl get pod
NAME READY STATUS RESTARTS AGE
ml-model-deployment-57f4778c74-zsfbb 0/1 ContainerCreating 0 33s

afzaa@AFZAALAHMEDKHAN MINGW64 ~/Desktop/ai/model
$ kubectl get pod
NAME READY STATUS RESTARTS AGE
ml-model-deployment-57f4778c74-zsfbb 1/1 Running 0 4m34s

afzaa@AFZAALAHMEDKHAN MINGW64 ~/Desktop/ai/model
$ kubectl get deploy
```

```
ml-model-deployment-57f4778c74-zsfbb 0/1 ContainerCreating 0 33s

afzaa@AFZAALAHMEDKHAN MINGW64 ~/Desktop/ai/model
$ kubectl get pod
NAME READY STATUS RESTARTS AGE
ml-model-deployment-57f4778c74-zsfbb 1/1 Running 0 4m34s

afzaa@AFZAALAHMEDKHAN MINGW64 ~/Desktop/ai/model
$ kubectl get deploy
NAME READY UP-TO-DATE AVAILABLE AGE
ml-model-deployment 1/1 1 1 4m42s

afzaa@AFZAALAHMEDKHAN MINGW64 ~/Desktop/ai/model
$ kubectl get svc
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
kubernetes ClusterIP 10.96.0.1 <none> 443/TCP 11m
ml-model-service NodePort 10.107.90.35 <none> 80:30007/TCP 7m53s

afzaa@AFZAALAHMEDKHAN MINGW64 ~/Desktop/ai/model
```

